Financial Performance Prediction Report

Final notebook, OOF validation, blend search, accounting postprocessing, and submission packaging.

1. Executive Summary

Best OOF blend mean R2: 0.857226605158

Selected accounting adjustment: none

Final experiment: M6_oof_blend

Best baseline / candidate chain: B4 -> M3/M4 -> M6 blend -> M7 accounting check

Submission rows/cols: 406 / 10

2. Environment and Raw Input Freeze

All automation was run in Conda environment QuantEnv with Python Python 3.10.20. Missing optional packages remain: xgboost, lightgbm, optuna, jupyter.

Raw file freeze manifest

name	size_bytes	sha256
train.csv	3455035	fc0d1fab1ed7b597cc900e1e724dc4847814fba7aa04dd722e21fbbb0d0f7049
test.csv	816771	a1003ff8a6f90a971e17d473781fff84621131f342f0e3d0fb2b0b59b886d7b6
sample_submission.csv	9291	874682cab231f9e48b6c038ac4859213ebf70453bf390ce94de31273da8dc1e6
data_dictionary.txt	7425	ba3697c21555558bb831d33da97b35b420558f2494530dc6763b85eaf30ca128
说明.docx	15737	c4e71f313214b7fe2fe1542d53a58ad75621e2f09875d501c18184e247819db6

3. Data Audit and EDA

Schema summary

dataset	rows	cols	id_unique	missing_cells	missing_rate	numeric_cols	object_cols	feature_cols	target_cols
train	1624	212	True	19283	0.056008	208	4	202	9
test	406	203	True	5791	0.070264	199	4	202	0
sample_submission	406	10	True	0	0.000000	10	0	0	0

Quarter-level missing rate summary

quarter	avg_missing_rate	median_missing_rate	max_missing_rate	min_missing_rate	observed_columns
Q1	0.013822	0.004926	0.147783	0.000000	17
Q2	0.016140	0.007389	0.156158	0.000000	17

Q3	0.031237	0.014778	0.193596	0.000000	17
Q4	0.018603	0.009852	0.143350	0.000000	17
Q5	0.021066	0.010345	0.142857	0.000000	17
Q6	0.028224	0.016256	0.155665	0.000000	17
Q7	0.119502	0.106897	0.254680	0.000000	17
Q8	0.038598	0.026601	0.153695	0.000000	17
Q9	0.082295	0.071921	0.181773	0.000000	17
Q10	0.128919	0.123153	0.200493	0.000000	17

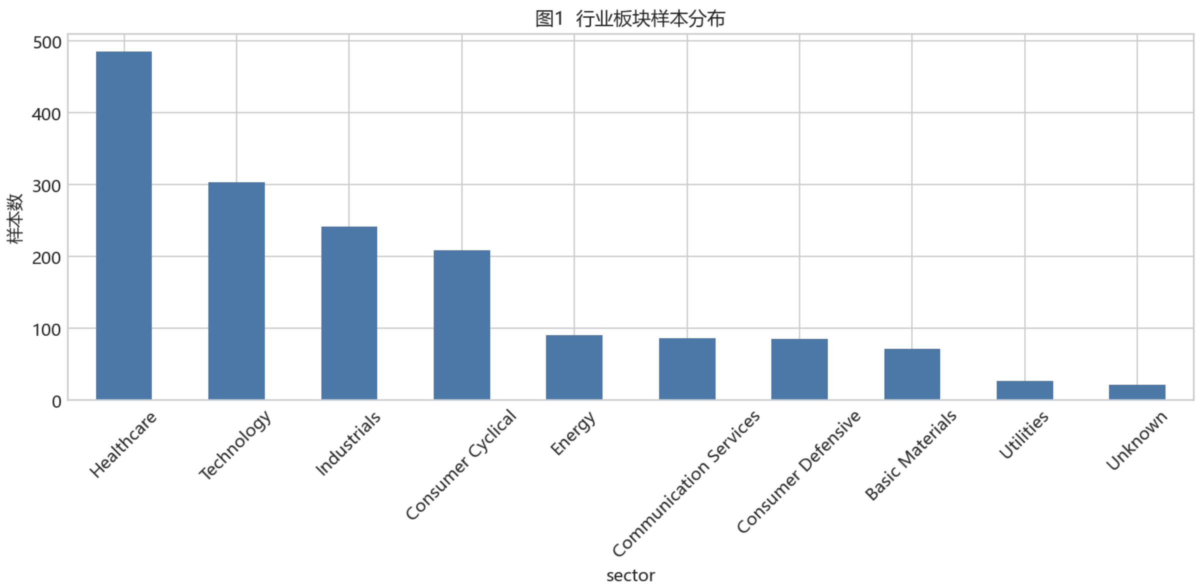

Figure 1. Sector sample distribution.

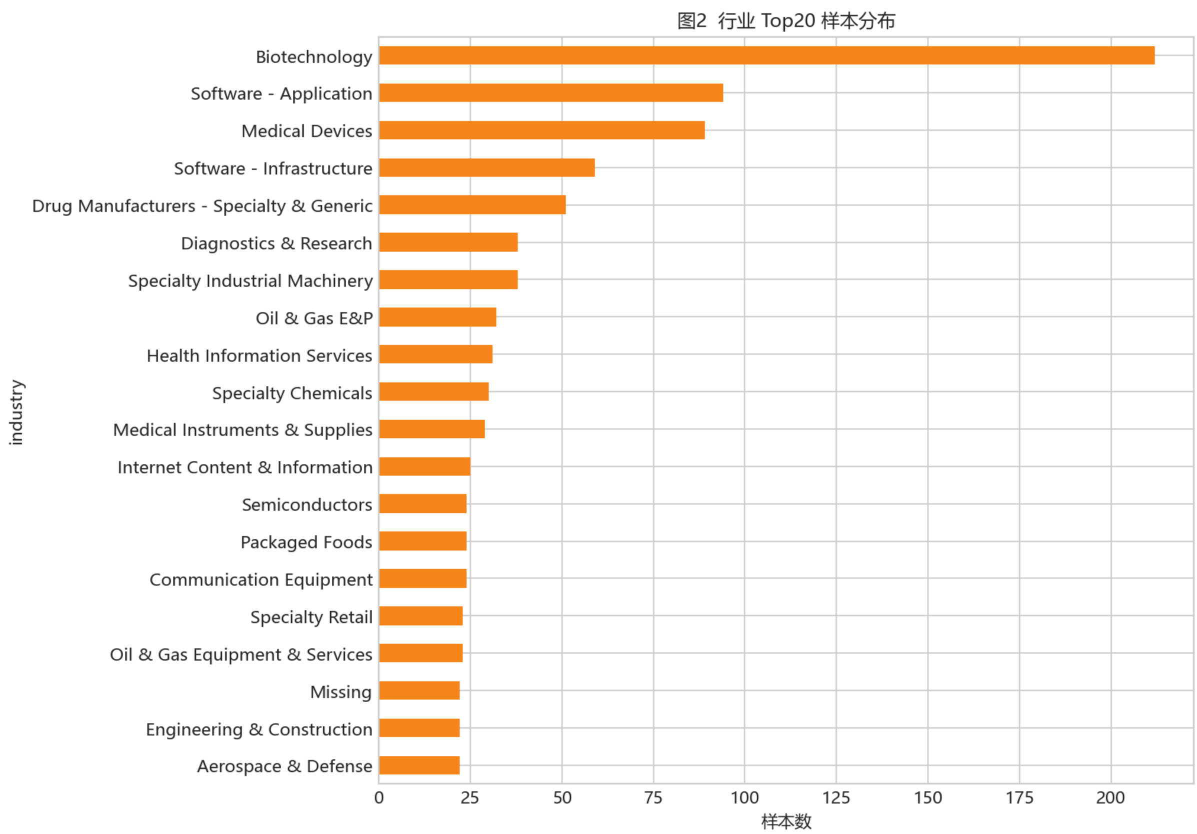

Figure 2. Top-20 industry sample counts.

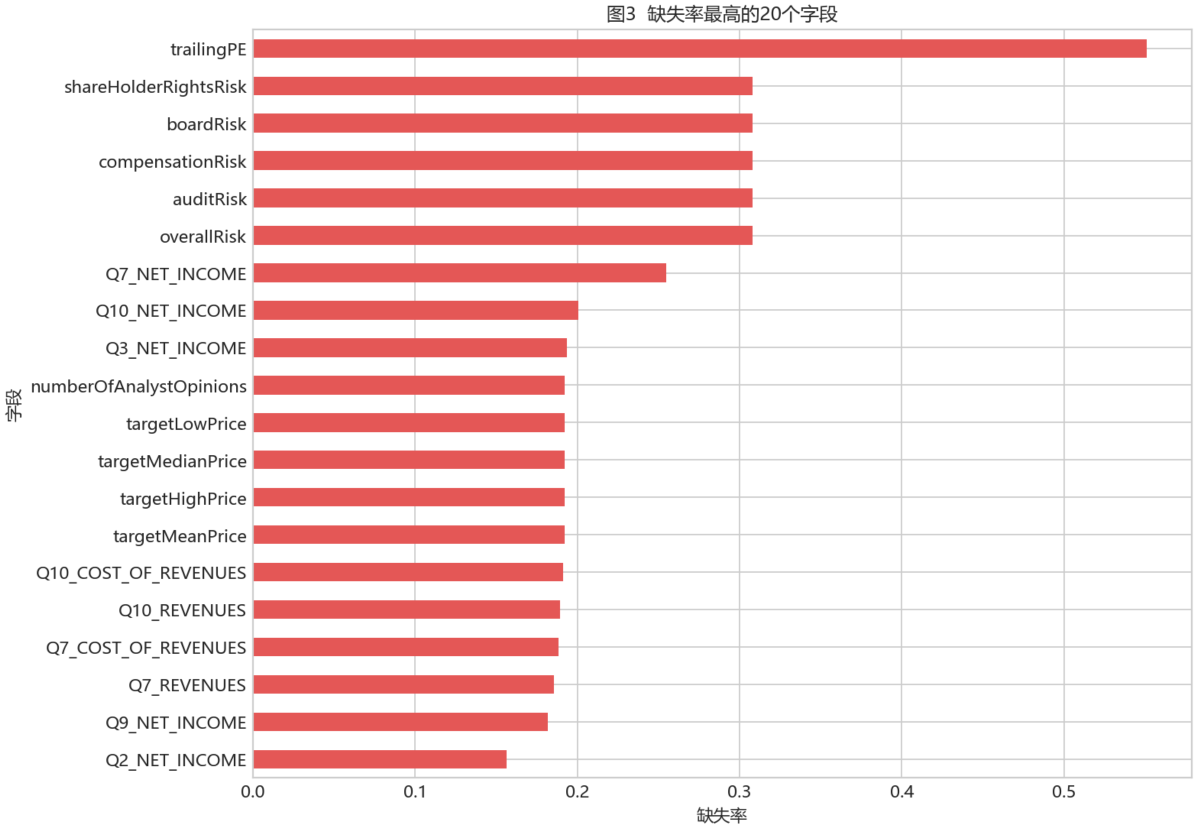

Figure 3. Top-20 missing-rate columns.

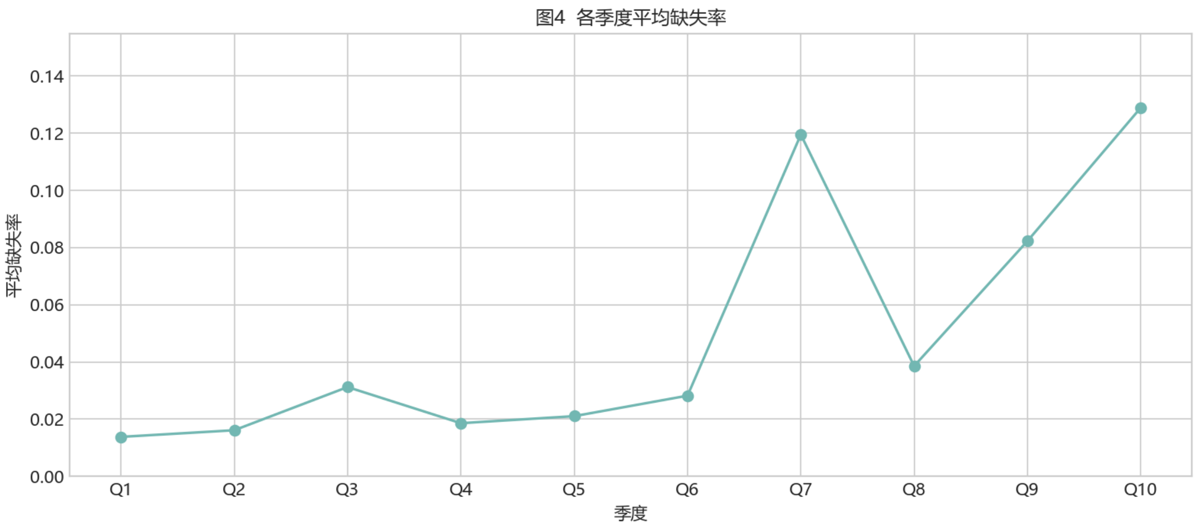

Figure 4. Mean missing rate by quarter.

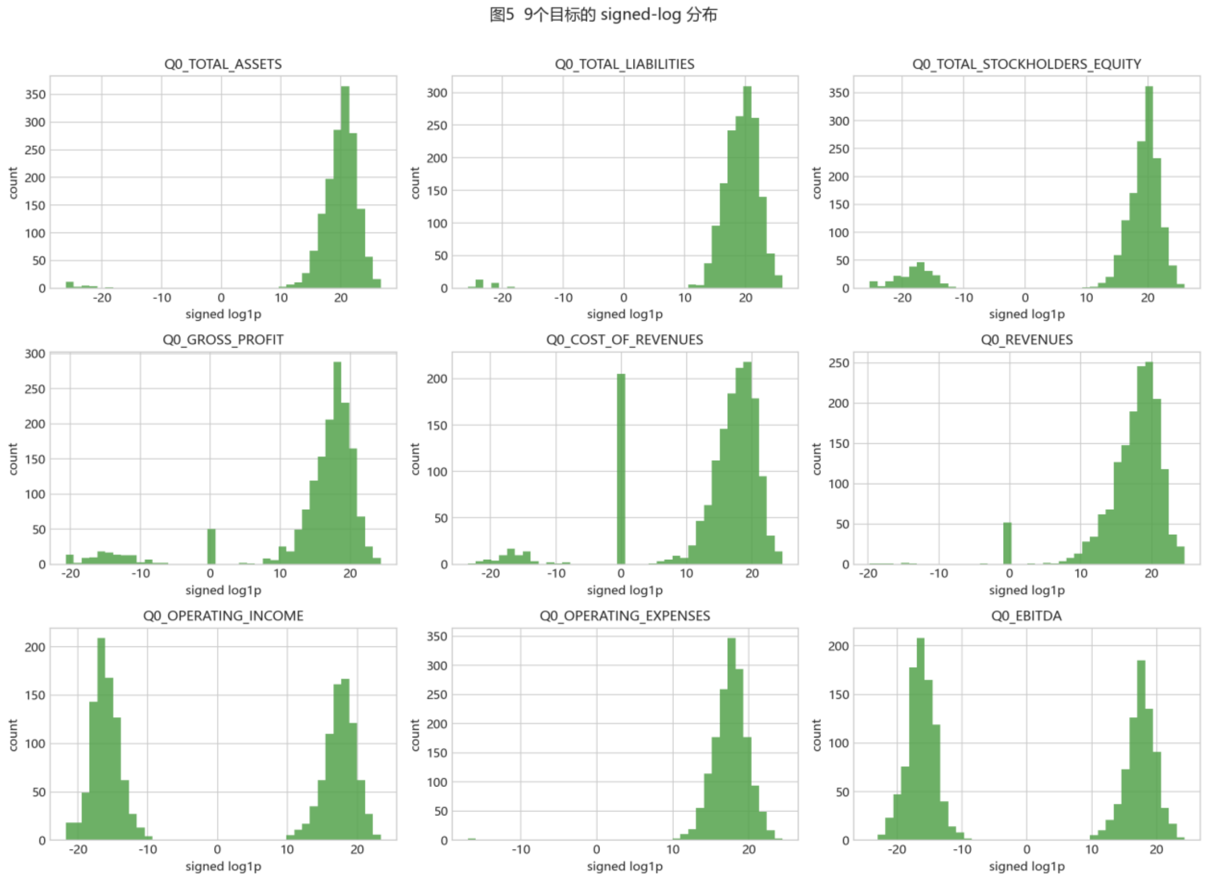

Figure 5. Signed-log target distributions.

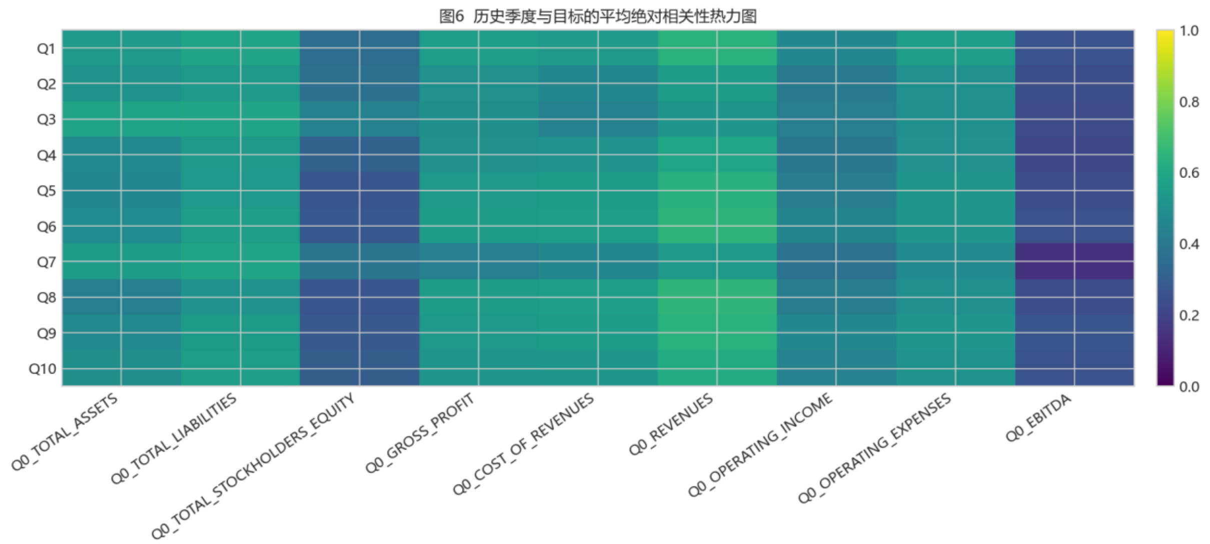

Figure 6. Lag correlation heatmap.

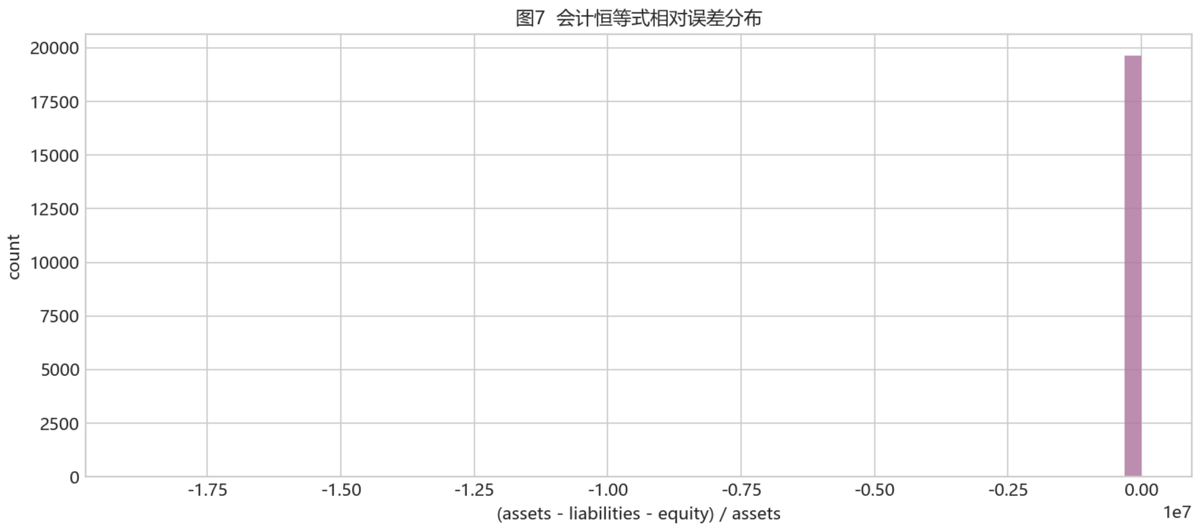

Figure 7. Accounting identity error distribution.

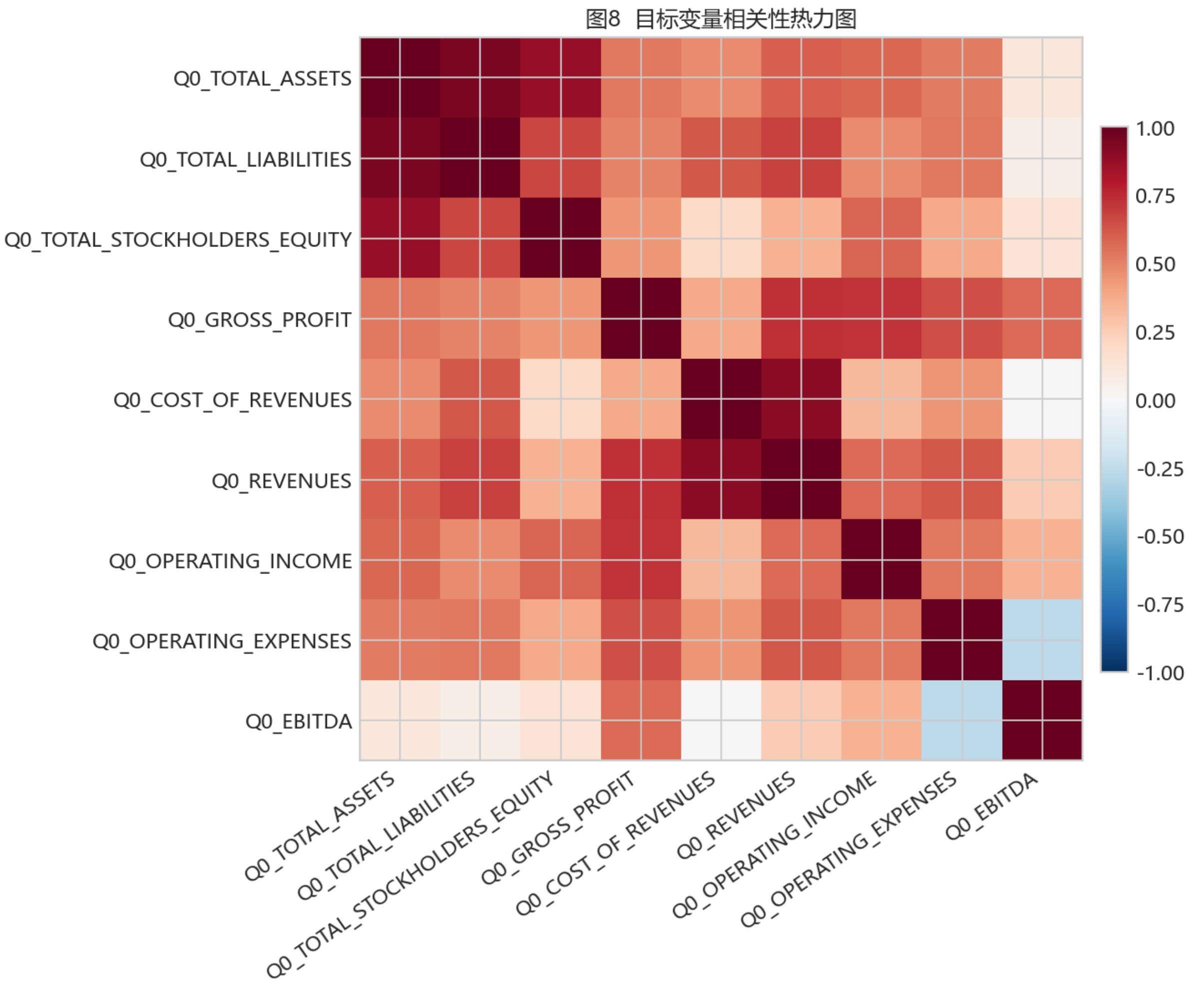

Figure 8. Target correlation heatmap.

4. Modeling and Validation

Selected experiment summary

experiment_id	model_name	target_strategy	mean_r2	fold_mean_r2_std	runtime_seconds

M7_accounting_postprocess	AccountingPostprocess	none	0.857227	0.049048	5.491926
M6_oof_blend	OOFWeightedBlend	per_target_oof_blend	0.857227	0.049048	5.491926
M4_catboost_residual_history_metadata_engineered	CatBoostRegressor	residual	0.825250	1.120254	330.111681
B4	B4_baseline	nan	0.783083	0.129085	2.092509
M1_ridge_history_raw	Ridge	direct	0.731390	0.190283	6.598222
M3_catboost_direct_history_metadata_engineered	CatBoostRegressor	direct	0.609040	0.433507	554.514580
M2_hgb_history_raw	HistGradientBoostingRegressor	direct	0.497172	0.114549	107.401618

Per-target OOF blend weights

target	members	weights	oof_r2	coarse_step	fine_step	fine_radius
Q0_TOTAL_ASSETS	B4,M3,M4	0.00,0.40,0.60	0.703489	0.050000	0.010000	0.050000
Q0_TOTAL_LIABILITIES	B4,M3,M4	0.00,0.00,1.00	0.911213	0.050000	0.010000	0.050000
Q0_TOTAL_STOCKHOLDERS_EQUITY	B4,M3,M4	0.00,1.00,0.00	0.400867	0.050000	0.010000	0.050000
Q0_GROSS_PROFIT	B4,M3,M4	0.00,0.00,1.00	0.984133	0.050000	0.010000	0.050000
Q0_COST_OF_REVENUES	B4,M3,M4	0.00,0.00,1.00	0.964448	0.050000	0.010000	0.050000
Q0_REVENUES	B4,M3,M4	0.30,0.00,0.70	0.972883	0.050000	0.010000	0.050000
Q0_OPERATING_INCOME	B4,M3,M4	0.00,0.00,1.00	0.826807	0.050000	0.010000	0.050000
Q0_OPERATING_EXPENSES	B4,M3,M4	0.97,0.03,0.00	0.987441	0.050000	0.010000	0.050000
Q0_EBITDA	B4,M3,M4	0.00,0.00,1.00	0.963759	0.050000	0.010000	0.050000

Accounting postprocessing candidates

adjustment	mean_r2	r2_Q0_TOTAL_ASSETS	r2_Q0_TOTAL_LIABILITIES	r2_Q0_TOTAL_STOCKHOLDERS_EQUITY	r2_Q0_GROSS_PROFIT	r2_Q0_COST_OF_REVENUES	r2_Q0_REVENUES	r2_Q0_OPERATING_INCOME	r2_Q0_OPERATING_EXPENSES	r2_Q0_EBITDA
none	0.857227	0.703489	0.911213	0.400867	0.984133	0.964448	0.972883	0.826807	0.987441	0.963759
balance_sheet_equity	0.836744	0.703489	0.911213	0.216526	0.984133	0.964448	0.972883	0.826807	0.987441	0.963759
gross_profit_identity	0.854654	0.703489	0.911213	0.400867	0.960980	0.964448	0.972883	0.826807	0.987441	0.963759

operating_income_identity	0.362620	0.703489	0.911213	0.400867	0.984133	0.964448	0.972883	-3.624653	0.987441	0.963759
income_statement_identity	0.342141	0.703489	0.911213	0.400867	0.960980	0.964448	0.972883	-3.785815	0.987441	0.963759
balance_and_income_identity	0.321658	0.703489	0.911213	0.216526	0.960980	0.964448	0.972883	-3.785815	0.987441	0.963759

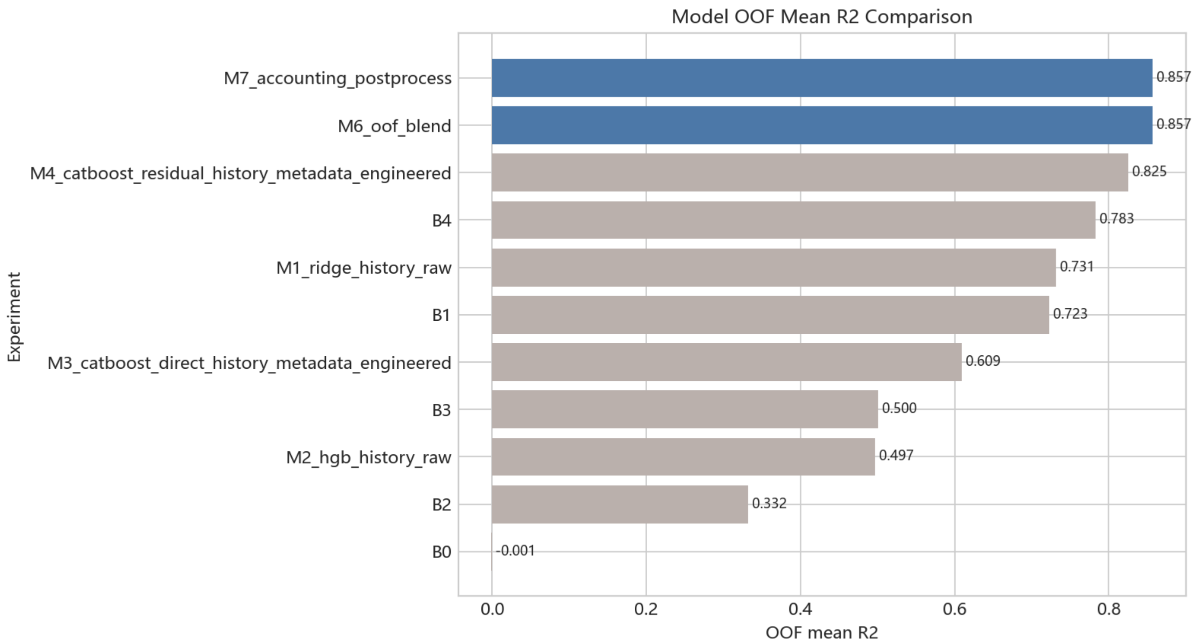

Figure 9. Model OOF mean R2 comparison.

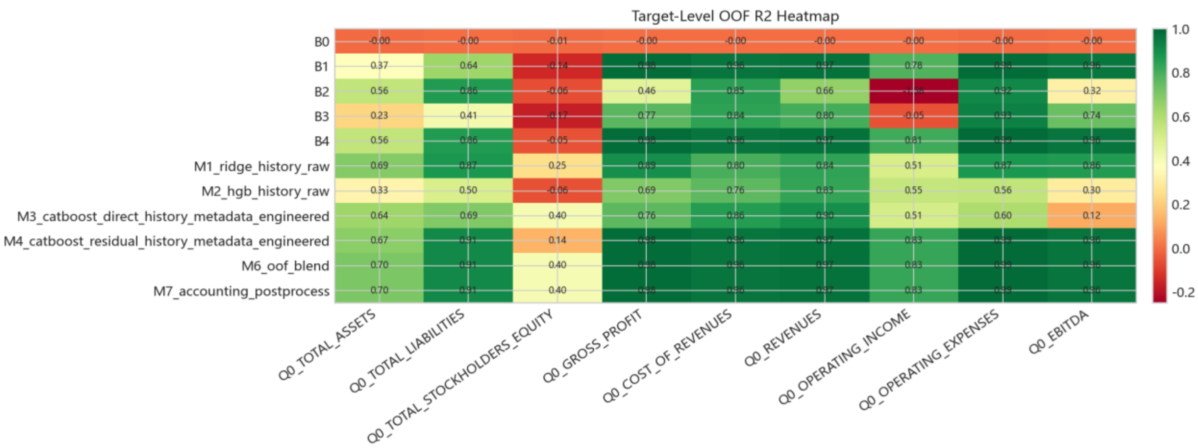

Figure 10. Target-level OOF R2 heatmap.

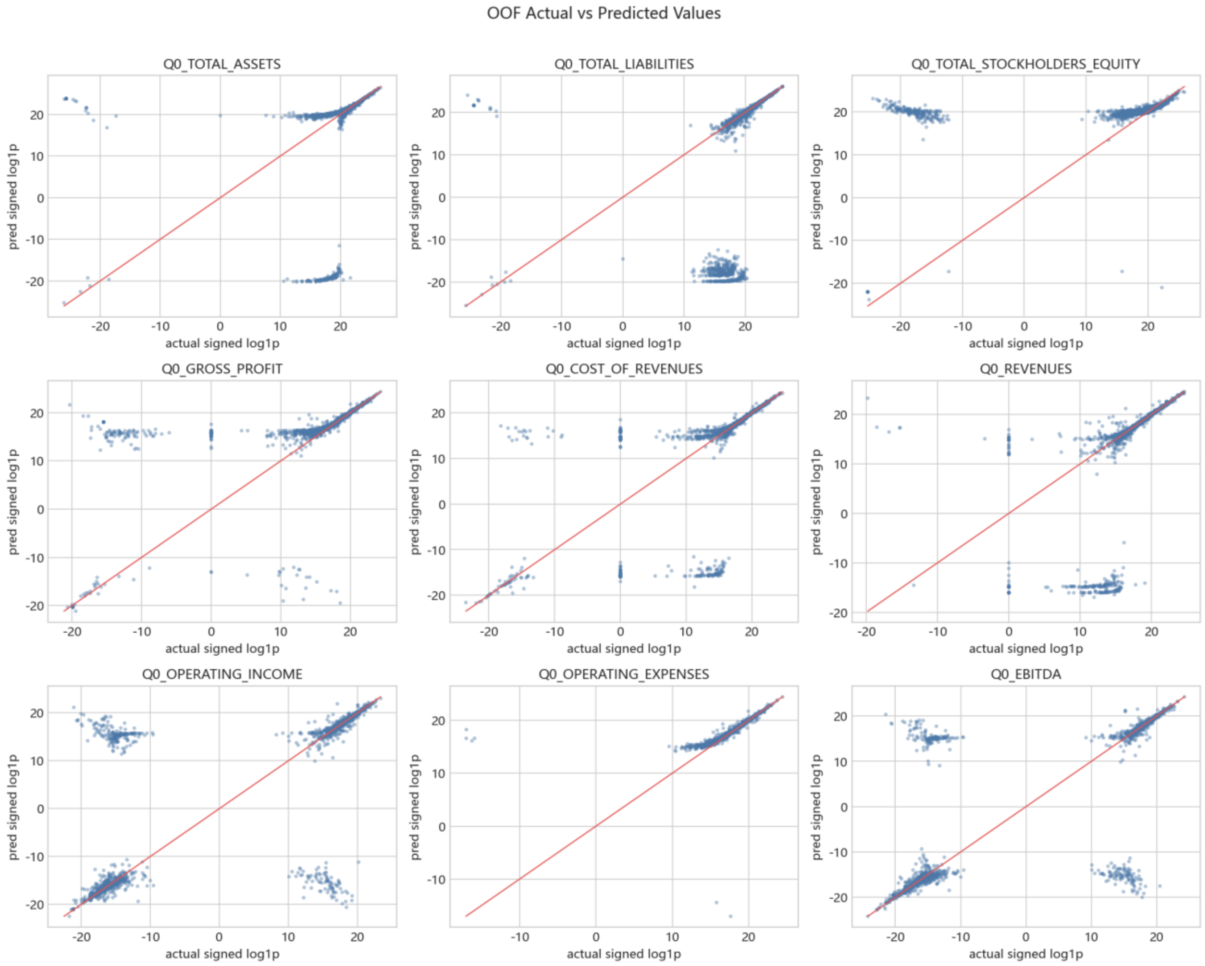

Figure 11. OOF actual-vs-predicted scatter plots.

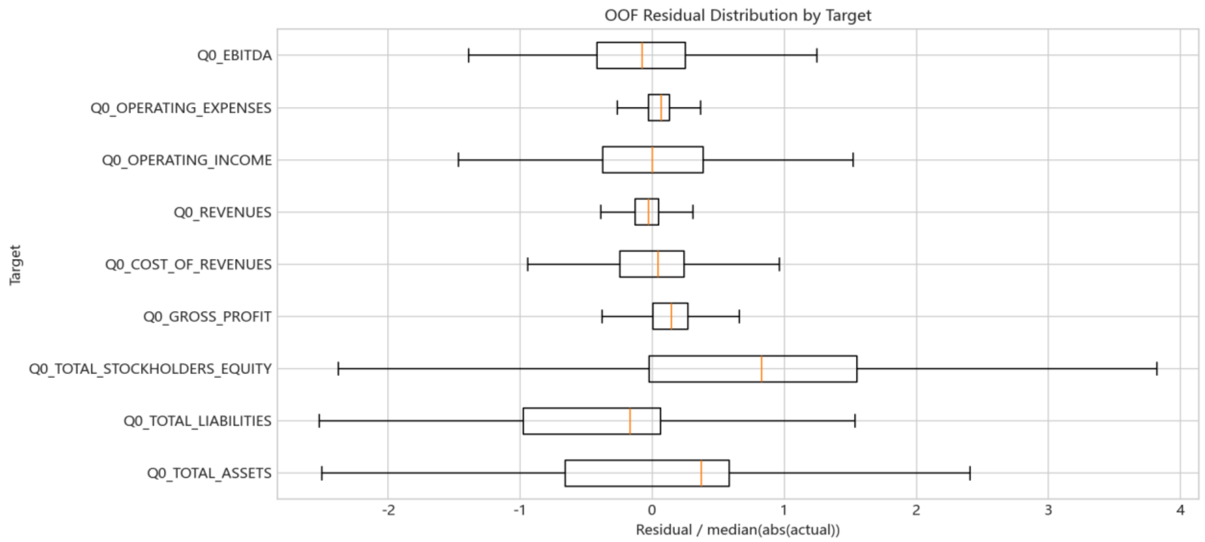

Figure 12. OOF residual distribution by target.

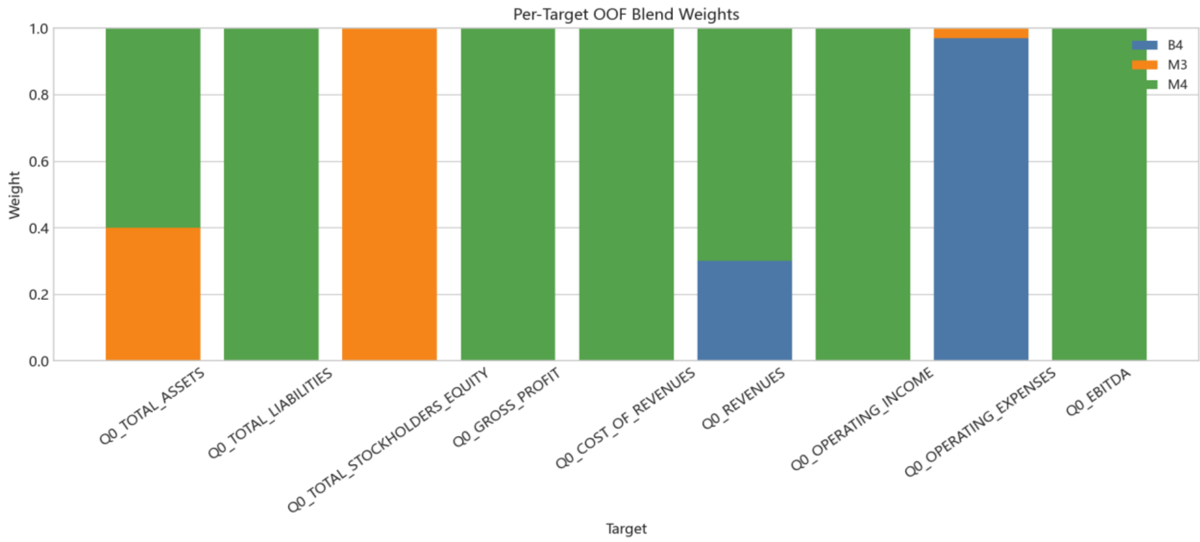

Figure 13. Per-target OOF blend weights.

5. Final Submission and Reproducibility

The final submission matches the sample template exactly with SHA256 c8c777fb68ca7d363e218b8ff09d4d7eb2e189f92084523779bc0de9c947e97f. The notebook and report are designed to be reproducible from the frozen raw inputs and saved results tables.

Key reproducibility command chain:

- conda run -n QuantEnv python scripts/check_environment.py
- conda run -n QuantEnv python scripts/bootstrap.py
- conda run -n QuantEnv python scripts/audit_data.py
- conda run -n QuantEnv python scripts/run_baselines.py
- conda run -n QuantEnv python scripts/run_sklearn_models.py
- conda run -n QuantEnv python scripts/run_catboost_models.py
- conda run -n QuantEnv python scripts/train_final.py
- conda run -n QuantEnv python scripts/build_final_figures.py
- conda run -n QuantEnv python scripts/build_notebook.py
- conda run -n QuantEnv python scripts/build_report.py
- conda run -n QuantEnv python scripts/export_report_pdf.py
- conda run -n QuantEnv python scripts/package_delivery.py
- conda run -n QuantEnv python scripts/validate_delivery.py